Small primitive strings with intransitive omissions

Human Compression project: bounded L044 component

We recover Proposition 5.10 and the sixteen literal coloured graphs of Table 2 in Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, physical pages 26–27. The source attributes this finite classification to a Magma computation. Our replacement uses the previously accepted prefix method with a new necessary orbit test and an exact correspondence with the source drawings. It is a required finite proof component, not a diagnostic. Proposition 5.11's subsequent attachment exclusions, Table 1's different rank maxima, and the full original remain outside this note.

Theorem 1. *Let $G \leq A_n$ be primitive on n points, $2 \leq n \leq 11$, with a string C-marking $(g_0, \ldots, g_{d-1})$ of nonidentity involutions and connected string diagram. Suppose every maximal marked omission is intransitive and $d \geq (n-1)/2$. Up to simultaneous permutation conjugacy and reversal of the entire string, its coloured permutation graph is exactly one of the sixteen graphs in L044 Table 2. There are 27 classes before taking reversal.*

Source rows	degree	rank	ordered classes	reversal classes
1: D_{10}	5	2	1	1
2–3: $A_5 \cong \mathrm{PSL}_2(5)$	6	3	3	2
4–9: A_9	9	4	11	6
10–13: A_{10}	10	5	8	4
14–16: A_{11}	11	6	4	3

Here an ordered class fixes every colour; only the separately stated full reversal identifies dual strings. No arbitrary reordering is used. The actual permutation isomorphism $A_5 \to \mathrm{PSL}_2(5)$ is the classical retained naming convention; the finite program identifies the degree-six group with A_5 explicitly.

Lemma 1 (Complete prefix enumeration)**.** *The programs* `census.g` *and* `run.g` *exhaust all markings in Theorem 1, modulo simultaneous conjugation.*

Proof. We retain the completeness of GAP's primitive-group catalogue in degrees 2 through 11. Every action is visited, including those with no output. A group containing an odd permutation is ineligible. If its involutions do not generate it, no involutory marking generates it. These exact tests leave seventeen actions.

For each remaining G, begin with the empty tuple. At a prefix $\mathbf{g}$ with group H, consider every involution outside H commuting with all but the last prefix generator and not commuting with the last. These are precisely the necessary next-letter conditions for an independent connected string. Conjugating by the stabilizer of the whole prefix in $N_{S_n}(G)$ preserves every condition; retaining one representative of each such orbit therefore loses no conjugacy class. The next stabilizer is its centralizer of the chosen involution.

Every accepted prefix is checked for independence. It must also satisfy

$$\langle g_0, \ldots, g_{l-1} \rangle \cap \langle g_1, \ldots, g_l \rangle = \langle g_1, \ldots, g_{l-1} \rangle.$$

This is used only as a necessary condition; no assertion that this single test proves all IP identities is needed. If a proper prefix is already transitive, then in any longer marking the omission of

the final generator contains that transitive prefix, which is forbidden. Such a prefix can be discarded.

Put $t = \lfloor n/2 \rfloor$. A prefix of length l needs at least $t - l$ further independent letters, so its index must be at least $2^{\max(0,t-l)}$. If $l > 0$, the letters after the next one generate a subgroup $T \leq C_G(H)$ of order at least 2^q, $q = \max(0, t - l - 1)$. Full IP in any eventual valid marking gives $T \cap H = 1$, hence $T \cap Z(H) = 1$ and

$$|C_G(H) : Z(H)| \geq 2^q.$$

Failure of either integer bound safely discards the prefix. The reused Whiston independent-family bound for a proper subgroup of S_n gives length at most $n - 2$; thus this is the complete upper search limit, not a rank guessed from the source table.

When $H = G$, the program checks the target rank and every maximal omission's orbit partition. It then constructs every marked parabolic and checks all 2^{2d} intersection equalities. Consequently every output is an eligible string C-marking, and the preceding necessary conditions ensure completeness. □

Proof of Theorem 1. The complete run gives the 27 ordered classes in the displayed table and no others. Their permutation tuples are written to the fresh correspondence receipt. The compulsory input `source_table2.json` independently transcribes every edge of each of the sixteen source drawings, including parallel edges with different colours; the coloured matchings reconstruct the literal involutions. Source vertices are numbered left to right on the upper path and then left to right below it.

For two such transitive coloured graphs, an isomorphism is uniquely determined by the image of any one vertex: propagate it along each colour, retaining loops at fixed points. The checker tries every possible image, rejects inconsistent or noninjective maps, and accepts only a complete bijection. Thus it checks actual simultaneous permutation conjugacy, not agreement of orders or Schläfli types. It does this both in the stated order and after reversing the full source tuple. Every computed marking matches one source row, every source row occurs, distinct source rows are inequivalent even under reversal, and the 27 computed ordered classes are pairwise inequivalent without reversal.

The degree-five group is generated by two involutions whose product has order five, and is D_{10}. GAP constructs an isomorphism of the degree-six group with A_5. The larger output groups are checked as the actual A_9, A_{10}, A_{11} in their natural actions. This verifies the group column as well as all literal markings. □

Proof and software boundary. The complete compulsory finite component consists of `census.g`, `run.g`, `check.py`, and `source_table2.json`. The Python checker performs the mathematical coloured-graph correspondence as well as launching GAP. GAP permutation-group arithmetic and its PrimGrp catalogue are retained software/classification inputs, with full unknown external proof cost. The earlier L047/common theory supplies Whiston's stated bound and the already reviewed prefix-method argument; these are paid dependencies, not a new theorem of this note. No earlier census output is used instead of the fresh run. The source page was directly viewed for all sixteen drawings. There is no new formalization, original admission, question solution, or priority claim.